

Conformal Symmetry in Physics

Simon Wood

The Australian National University
Department of Theoretical Physics

21 May 2014

Outline

- 1 What is symmetry?
- 2 Symmetry in physics
- 3 Conservation laws
- 4 Conformal symmetry

What is symmetry?

Shapes with equal symmetry

8 independent symmetry transformations: rotations and reflections

Shapes with different symmetry

8 independent symmetry transformations: no reflections, only rotations
on the right

How does one distinguish symmetries?

Symmetries form an algebraic structure called a *group*.

Group axioms

A group is a set G together with multiplication rules $G \times G \rightarrow G$, such that:

- There exists an identity element 1 such that $1 \cdot g = g, \forall g \in G$.
- For each element g there exists an inverse element g^{-1} such that $g \cdot g^{-1} = 1$.
- Multiplication is associative $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$.

How does one distinguish symmetries?

Two groups or symmetries are different if the multiplication rules are different.

Symmetry group of square or four pointed star:

$\times$	1	r_4	r_4^2	r_4^3	s	sr_4	sr_4^2	sr_4^3
1	1	r_4	r_4^2	r_4^3	s	sr_4	sr_4^2	sr_4^3
r_4	r_4	r_4^2	r_4^3	1	sr_4^3	s	sr_4	sr_4^2
r_4^2	r_4^2	r_4^3	1	r_4	sr_4^2	sr_4^3	s	sr_4
r_4^3	r_4^3	1	r_4	r_4^2	sr_4	sr_4^2	sr_4^3	s
s	s	sr_4	sr_4^2	sr_4^3	1	r_4	r_4^2	r_4^3
sr_4	sr_4	sr_4^2	sr_4^3	s	r_4^3	1	r_4	r_4^2
sr_4^2	sr_4^2	sr_4^3	s	sr_4	r_4^2	r_4^3	1	r_4
sr_4^3	sr_4^3	s	sr_4	sr_4^2	r_4	r_4^2	r_4^3	1

Where r_4 is a quarter counter clockwise rotation and s is the reflection about the vertical axis.

Symmetry group of twisted eight pointed star:

$$r_8^m \cdot r_8^n = r_8^{m+n}, \quad r_8^8 = 1,$$

where r_8 is an eighth counter clockwise rotation.

Symmetry in physics

In physics, a symmetry is a transformation that does not change equations of motion.

Equivalently: Symmetry transformations map solutions of equations of motion to other solutions.

Example: The harmonic oscillator (a spring)

The equation of motion is given by Newton's second law:

$$m\ddot{x}(t) = -kx(t)$$

Example: The harmonic oscillator (a spring)

The equation of motion is given by Newton's second law:

$$m\ddot{x}(t + T) = -kx(t + T)$$

This equation is time translation symmetric.

A selection of great achievements of symmetry in physics

- Two body problem
 - ▶ classical $\longrightarrow$ planetary motion
 - ▶ quantum $\longrightarrow$ hydrogen atom
- Symmetries of Newtonian mechanics $\neq$ Symmetries of electrodynamics $\longrightarrow$ special relativity
- Independence of laws of physics on the choice of coordinates $\longrightarrow$ general relativity
- Particle physics: symmetries among the forces of nature

Conservation laws

Arguably the most important theorem regarding symmetries in physics is Emmy Noether's theorem which relates symmetries to conservation laws.

Noether's Theorem

Any continuous (differentiable) symmetry of a physical system has a corresponding conservation law.

Example: Conservation of energy

$$m\ddot{x}(t) = -kx(t) = -\frac{d}{dx}U(x(t))$$

Example: Conservation of energy

$$m\ddot{x}(t) = -kx(t) = -\frac{d}{dx}U(x(t))$$

$$m\ddot{x}(t) + \frac{d}{dx}U(x(t)) = 0$$

Example: Conservation of energy

$$m\ddot{x}(t) = -kx(t) = -\frac{d}{dx}U(x(t)) \quad m\dot{x}(t)\ddot{x}(t) + \dot{x}(t)\frac{d}{dx}U(x(t)) = 0$$

$$m\ddot{x}(t) + \frac{d}{dx}U(x(t)) = 0$$

Example: Conservation of energy

$$m\ddot{x}(t) = -kx(t) = -\frac{d}{dx}U(x(t)) \quad m\dot{x}(t)\ddot{x}(t) + \dot{x}(t)\frac{d}{dx}U(x(t)) = 0$$

$$m\ddot{x}(t) + \frac{d}{dx}U(x(t)) = 0 \quad \frac{d}{dt}\left(\frac{m}{2}\dot{x}(t)^2 + U(x(t))\right) = 0$$

$$\text{Kinetic energy} = \frac{m}{2}\dot{x}(t)^2$$

Conformal symmetry

Conformal transformation

A conformal transformation is a transformation that preserves angles.

Conformal symmetry

Conformal transformation

A conformal transformation is a transformation that preserves angles.

In 2 dimensions there are infinitely many independent (local) conformal transformations.

Examples of conformal transformations

Translation

Examples of conformal transformations

Rotation

Examples of conformal transformations

Scaling transformation (zoom in or out)

Examples of conformal transformations

Special conformal transformation

Applications of conformal symmetry

- Particle physics (limits and toy models)
- String theory (my motivation for understanding conformal symmetry)
- Many body physics a.k.a. statistical mechanics

Real world examples of conformal symmetry

For the remainder of the talk I will present two examples with explicit scaling symmetry:

- The Ising model
- Critical opalescence

The Ising model

The Ising model: a simple model to describe the magnetism of iron. Think of a large grid of little magnets (iron atoms).

- Each atom can point in two directions.
Up is white. Down is black.
- Each magnet will try to point in the same direction as its neighbours.

Magnetisation diagram

Source: Simplified version of image found on Wikipedia

Ising model simulation

The following video was made by Douglas Ashton and can be found on YouTube.

Title: The Renormalisation Group

Critical opalescence

Critical opalescence is a phenomenon that occurs for many pure substances such as water or CO_2 .

These substances are clear at almost all temperatures and pressures and become opaque at a special critical temperature and pressure.

Pressure-Temperature diagram

Source: Simplified version of image found on Wikipedia

Critical opalescence video

The following video of CO_2 at its critical point can be found on YouTube.
Title: Supercritical fluids

How critical opalescence can be understood from scaling symmetry

How critical opalescence can be understood from scaling symmetry

How critical opalescence can be understood from scaling symmetry

The End

Thank you for your attention!